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1

COMP

Calcium homeostasis

You should be able to: Trace the response to falling plasma calcium through parathyroid hormone, calcitriol, and the actions on bone, kidney, and intestine, and state the role of calcitonin.

PICK ONE AND DO THAT ONE

☐ **A** Draw plasma calcium as a bar with the normal band marked, about 8.5 to 10.5 mg/dL, and drop the level below it. Draw the parathyroid glands releasing PTH, then three arrows out to bone, kidney, and intestine. On the kidney arrow show calcium reabsorption up, phosphate reabsorption down, and the step that activates calcitriol. Route the intestinal arrow through calcitriol. One line: name the target PTH cannot reach without calcitriol, and say why.

☐ **B** Draw one calcium scale with two arrows off it, one for calcium falling and one for calcium rising, and put the hormone that answers each beside it, PTH from the parathyroids on one and calcitonin from the thyroid C cells on the other. Under each, draw its effect on the osteoclast. Then draw plasma calcium and nerve excitability after the parathyroids are removed in thyroid surgery. One line: say which of the two arms you could lose and barely notice.

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2

COMP

Kidney functions

You should be able to: List the homeostatic functions of the kidney including water and electrolyte balance, acid base regulation, waste excretion, blood pressure control, and hormone production.

PICK ONE AND DO THAT ONE

☐ **A** Draw one kidney large in the middle of the page with six arrows leaving it, one per job: water balance, electrolyte balance, acid base balance, waste excretion, blood pressure control, and hormone production. At the end of each arrow draw the actual thing moved or made, a water molecule, a sodium ion, a hydrogen ion, a urea molecule, renin, erythropoietin. One line: pick the job whose failure you would notice first in a patient, and name the sign.

☐ **B** Draw a body outline for a patient whose kidneys have failed and mark the consequences on it rather than listing them: where fluid collects, what plasma potassium does, what arterial pH does, what builds up in the blood, and what happens to the red cell count. Beside each mark write which kidney job was lost to cause it. One line: dialysis replaces some of these and not others, so name one it does not replace.

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3

Nephron structure and function

COMP

You should be able to: Match each nephron segment to its dominant transport function and distinguish a cortical from a juxtamedullary nephron by capability.

PICK ONE AND DO THAT ONE

☐ **A** Draw one nephron stretched left to right from glomerulus to collecting duct, with cortex and medulla as bands behind it. Label Bowman's capsule, proximal tubule, descending limb, thick ascending limb, distal tubule, and collecting duct. Under each segment write its dominant transport job in a few words. One line: name the segment that stays impermeable to water no matter what the hormones are doing, and say what that buys the kidney.

☐ **B** Draw a wedge of kidney with cortex and medulla, then draw a cortical nephron and a juxtamedullary nephron inside it at true relative depth and loop length. Add peritubular capillaries to one and vasa recta to the other. Mark roughly what fraction of nephrons is each type. One line: say which type you need to make urine more concentrated than plasma, and what would happen if you had only the other kind.

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4

COMP

Three renal processes

You should be able to: Define filtration, reabsorption, secretion, and excretion and write the equation that relates them for any solute.

PICK ONE AND DO THAT ONE

☐ **A** Draw a nephron with its capillaries and put the three arrows in the right places and directions: filtration at the glomerulus into the capsule, reabsorption from tubule back to blood, secretion from blood into tubule. Write the equation for excretion in terms of the other three beside the drawing. One line: pick a substance that is filtered and then almost entirely reabsorbed, and say what its excretion rate looks like.

☐ **B** Draw the same nephron three times in a row, once for glucose, once for creatinine, and once for a secreted drug such as penicillin. On each copy draw only the arrows that actually happen for that substance, with arrow thickness showing how much. One line: say which of the three leaves in the urine at a higher rate than it was filtered, and what that proves about a step.

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5

Filtration membrane and selectivity

COMP

You should be able to: Describe the three layers of the filtration membrane and predict which plasma components appear in the filtrate based on size and charge.

PICK ONE AND DO THAT ONE

☐ **A** Draw a cross section through the filtration membrane with blood on one side and capsular space on the other, showing all three layers: fenestrated endothelium, basement membrane, and podocyte foot processes with slits between them. Mark the negative charge on the basement membrane. Then draw four things arriving, water, glucose, albumin, and a red blood cell, and show how far each gets. One line: say why albumin is held back for two separate reasons, not one.

☐ **B** Draw two filtration membranes side by side, a normal one and one whose basement membrane has lost its negative charge. Send the same set of plasma components at both: water, sodium, glucose, a small protein, albumin, and a red cell. Show what crosses in each. Beside the damaged one write the urinalysis result you expect. One line: say what weeks of this does to plasma colloid osmotic pressure, and what that does to tissue fluid.

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6

COMP

Net filtration pressure and GFR

You should be able to: Calculate net filtration pressure from glomerular hydrostatic pressure, capsular pressure, and colloid osmotic pressure and predict the effect of each change on glomerular filtration rate.

PICK ONE AND DO THAT ONE

☐ **A** Draw a glomerular capillary with three arrows drawn to scale: glomerular hydrostatic pressure about 55 mmHg pushing out, capsular hydrostatic pressure about 15 mmHg pushing in, and blood colloid osmotic pressure about 30 mmHg pulling in. Work the net filtration pressure out on the page and mark normal GFR near 125 mL/min beside it. One line: say which single one of those three pressures the body actually adjusts, and how it does it.

☐ **B** Draw the same three pressure arrows three more times, once for a stone blocking the ureter, once for liver failure with low plasma protein, and once for hemorrhage dropping arterial pressure. In each, redraw only the arrow that changes and write the new net filtration pressure and what GFR does. One line: two of these three move GFR the same direction, so name the odd one out and say why it goes the other way.

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COMP

Regulation of GFR

You should be able to: Compare myogenic autoregulation, tubuloglomerular feedback, sympathetic control, and hormonal control and predict the effect of afferent and efferent arteriolar constriction on filtration rate.

PICK ONE AND DO THAT ONE

☐ **A** Draw a glomerulus with its afferent and efferent arterioles clearly distinct, then draw four copies: afferent constricted, afferent dilated, efferent constricted, efferent dilated. On each mark what happened to glomerular hydrostatic pressure and to GFR with an up or down arrow. One line: two of the four raise GFR, so name them, and say which of those two cuts total renal blood flow at the same time.

☐ **B** Draw an arterial pressure trace that dips and then climbs above normal, and off it draw three controls responding: the myogenic response in afferent smooth muscle, tubuloglomerular feedback from the macula densa back to the afferent arteriole, and sympathetic outflow during hemorrhage. Show what each does to arteriole diameter. One line: say which of these gives up when arterial pressure falls far enough, and what takes over when it does.

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8

COMP

Renal clearance

You should be able to: Calculate the clearance of a substance and use inulin and creatinine clearance to estimate glomerular filtration rate.

PICK ONE AND DO THAT ONE

☐ **A** Draw the clearance equation as a picture rather than symbols: a urine collection with its concentration and its volume per minute, an arrow to a plasma sample with its concentration, and the volume of plasma cleared per minute coming out. Then draw a nephron with inulin passing through, filtered, not reabsorbed, not secreted. Write inulin's clearance next to normal GFR near 125 mL/min. One line: say why inulin has to be infused while creatinine does not.

☐ **B** Draw two nephrons side by side, one handling inulin and one handling creatinine, with every arrow that actually happens on each. Creatinine gets a small secretion arrow. Under each, draw a bar for its measured clearance against a marked line at the true GFR. One line: say which direction creatinine clearance is wrong as a GFR estimate, and whether that error makes a failing kidney look better or worse than it is.

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9

Clearance and handling inference

COMP

You should be able to: Use a clearance value relative to glomerular filtration rate to determine whether a substance is net reabsorbed or net secreted.

PICK ONE AND DO THAT ONE

☐ **A** Draw a number line for clearance in mL/min with GFR marked at about 125. Place inulin, glucose, sodium, creatinine, and PAH on it. Above the line write net reabsorbed, neither, or net secreted over the right zones. Beside each substance draw the single nephron arrow that puts it where you placed it. One line: pick the substance whose position moves in a diabetic with plasma glucose near 300 mg/dL, and say which way it moves.

☐ **B** You are handed three unknowns with clearances of 0, 125, and 400 mL/min in a person with normal GFR. Draw a nephron for each and put in only the arrows those numbers force. Beside each, name a real substance that behaves that way. One line: a clearance of 60 mL/min is ambiguous, so give the two different tubular stories that could both produce it.

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10

COMP

Proximal tubule reabsorption

You should be able to: Explain how the sodium gradient drives reabsorption of glucose, amino acids, and water in the proximal tubule and state why reabsorption there is described as isosmotic.

PICK ONE AND DO THAT ONE

☐ **A** Draw one proximal tubule cell with lumen on one side and blood on the other. Put the sodium potassium ATPase on the basolateral membrane and show the sodium gradient it creates. On the apical membrane draw the cotransporters that ride that gradient, sodium with glucose and sodium with an amino acid. Then draw water following. One line: say what happens to glucose reabsorption within minutes if you poison that pump, given that glucose never touches it.

☐ **B** Draw the proximal tubule as a tube and plot two things along its length: the volume of fluid remaining in it, and the osmolarity of that fluid starting near 290 mOsm/L. Show roughly two thirds of the water and sodium gone by the end and the osmolarity line flat. Draw where the water left. One line: say what would have to be true of the wall for that osmolarity line to climb instead, and name a segment where it does.

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11**Transport maximum and renal threshold**

COMP

You should be able to: Distinguish transport maximum from renal threshold and explain the appearance of glucose in urine when plasma glucose is elevated.

PICK ONE AND DO THAT ONE

☐ **A** Draw one set of axes with plasma glucose on the x and rate on the y, and plot three curves: filtered load rising as a straight line, reabsorption rising then flattening at the transport maximum near 375 mg/min, and excretion flat at zero then lifting off. Mark the renal threshold near 180 mg/dL where excretion leaves zero. One line: say why the threshold sits below the plasma level at which every transporter is finally saturated.

☐ **B** Draw the proximal tubule wall with a fixed number of glucose carriers on it, say six, and draw three cases passing by: normal plasma glucose, plasma glucose near 200 mg/dL, and plasma glucose near 400 mg/dL. Show carriers occupied, glucose left in the lumen, and water dragged along with it. One line: say why the patient in the third panel is thirsty and urinating heavily, and name the force pulling that water.

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12**Tubular secretion**

COMP

You should be able to: State the purpose of tubular secretion and name substances secreted including hydrogen ion, potassium, and organic anions such as drugs.

PICK ONE AND DO THAT ONE

☐ **A** Draw a distal nephron with its peritubular capillary alongside and four secretion arrows crossing from blood into lumen: hydrogen ion, potassium, ammonium, and an organic anion such as penicillin. Beside each arrow write why that substance is secreted rather than simply left in the filtrate. One line: say what secretion can do for a drug bound to plasma protein that filtration cannot do at all.

☐ **B** Pick a secreted drug that is not penicillin and draw the nephron twice for it. In the first copy show its route through filtration and secretion into the urine. In the second, block the secretion transporter and draw a small graph of what its plasma level does over the next few hours. One line: say what dose change you would make for that second patient, and in which direction.

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13**Countercurrent mechanism**

COMP

You should be able to: Explain how the countercurrent multiplier of the loop of Henle and the countercurrent exchanger of the vasa recta build and preserve the medullary osmotic gradient.

PICK ONE AND DO THAT ONE

☐ **A** Draw a loop of Henle in the medulla with an osmolarity scale down the side, about 300 mOsm/L at the cortex to about 1200 mOsm/L at the papilla. Mark the descending limb as permeable to water and not solute, and the thick ascending limb as pumping sodium chloride out while staying water impermeable. Write fluid osmolarity at the top, the bend, and the exit. One line: fluid leaving the ascending limb is more dilute than plasma, so say what that buys.

☐ **B** Draw the vasa recta as a hairpin running alongside the loop, blood flowing down and back up through the gradient. Show water and solute crossing at each level, and mark the osmolarity of blood entering and leaving. Then draw what the medullary gradient would look like if that vessel ran straight through instead of hairpinning. One line: say which structure is the multiplier and which is only the exchanger, and what the difference between those two words is.

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14

Antidiuretic hormone and urine concentration

COMP

You should be able to: Trace the response to dehydration from osmoreceptor firing through antidiuretic hormone release to aquaporin insertion and concentrated urine.

PICK ONE AND DO THAT ONE

☐ **A** Draw the dehydration chain as boxes and arrows: plasma osmolarity climbing above about 290 mOsm/L, hypothalamic osmoreceptors firing, ADH released from the posterior pituitary, and the collecting duct cell responding. Draw that last cell in detail, with aquaporins moving from vesicles into the apical membrane and water following. Show the urine leaving, small and concentrated. One line: name the second stimulus that can release ADH, and give a case where it overrides osmolarity.

☐ **B** Draw one collecting duct twice running down through the medullary gradient, once with maximal ADH and once with none. On each draw the aquaporins present or absent, the water arrows, and the volume and osmolarity of urine leaving the bottom. Use real urine osmolarity numbers if you can. One line: the medullary gradient is identical in both drawings, so say exactly what changed and where it changed.

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15

Renin angiotensin aldosterone system

COMP

You should be able to: Trace the renin angiotensin aldosterone pathway from the stimulus for renin release to the effects of angiotensin II and aldosterone on vessels, tubules, and blood pressure.

PICK ONE AND DO THAT ONE

☐ **A** Draw the chain starting at the stimulus, low blood pressure or low sodium chloride sensed at the macula densa, then granular cells releasing renin, angiotensinogen from the liver becoming angiotensin I, and ACE in the lung making angiotensin II. From angiotensin II draw four arrows to arterioles, adrenal cortex, hypothalamus, and proximal tubule, with the effect at each. One line: name the step a common blood pressure drug blocks, and say what happens downstream of it.

☐ **B** Draw a collecting duct principal cell with aldosterone arriving, entering the nucleus, and increasing two proteins: the apical sodium channel and the basolateral sodium potassium pump. Show sodium moving, potassium moving, and water following. Then draw the same cell in someone whose adrenal cortex has failed. One line: predict that person's blood pressure and plasma potassium, and say which of the two would kill them first.

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16**Natriuretic peptides**

COMP

You should be able to: Explain how atrial natriuretic peptide opposes the renin angiotensin aldosterone system and state the stimulus for its release.

PICK ONE AND DO THAT ONE

☐ **A** Draw a heart with the atria stretched by high blood volume, releasing atrial natriuretic peptide. From it draw arrows to four places and show the effect at each: the afferent and efferent arterioles, the collecting duct sodium channels, the granular cells making renin, and the adrenal cortex making aldosterone. Show what ends up in the urine. One line: state the release stimulus in one phrase, and say why the atrium is a sensible place to sense it.

☐ **B** Draw two columns, renin angiotensin aldosterone on one side and atrial natriuretic peptide on the other, and run four rows across both: stimulus, effect on arteriole tone, effect on sodium reabsorption, effect on blood volume. Draw every cell as a small picture rather than a sentence. One line: a person in heart failure has both running high at once, so say which one is winning and what that does to their fluid.

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17

COMP

Potassium handling

You should be able to: Explain how aldosterone and plasma potassium regulate potassium secretion in the collecting duct and predict the effect of a potassium wasting diuretic.

PICK ONE AND DO THAT ONE

☐ **A** Draw a collecting duct principal cell with the sodium potassium pump on the blood side and the potassium channel on the lumen side. Draw it three times: normal plasma potassium, high plasma potassium, and high aldosterone. Show how much potassium is secreted in each. One line: name the loop that lets high plasma potassium raise aldosterone as well as acting on this cell directly, and say why that makes the response larger.

☐ **B** Draw a nephron and mark where a loop diuretic blocks sodium reabsorption in the thick ascending limb. Then draw the collecting duct downstream and show what the extra sodium delivery and faster flow do to potassium secretion there. Plot plasma potassium across a week on the drug. One line: say what you would see on that patient's ECG by the end of the week, and what you would add to the prescription.

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18

COMP

Micturition

You should be able to: Trace the micturition reflex and explain how the internal and external sphincters allow voluntary control.

PICK ONE AND DO THAT ONE

☐ **A** Draw the bladder filling, with stretch receptors in the wall, the sensory path to the sacral spinal cord, the parasympathetic motor path back to the detrusor muscle, and both sphincters. Mark the internal sphincter as smooth muscle and involuntary, the external as skeletal muscle on the pudendal nerve. Show the state of each during filling and during voiding. One line: say which sphincter a toddler has to learn to control, and why the other needs no training.

☐ **B** Draw the reflex twice, once intact with the descending path from the pons included, and once with the spinal cord cut above the sacral segments. In the second, show which parts of the loop still work and which are gone, then draw the bladder behavior that results. One line: say why that patient still empties the bladder but cannot choose when, and name the piece that was lost.

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19

COMP

Urinalysis

You should be able to: Perform a urinalysis measuring specific gravity, pH, glucose, protein, ketones, and blood and relate any abnormal finding to the renal process that failed.

PICK ONE AND DO THAT ONE

☐ **A** Draw a dipstick with its pads in a row and write the normal result on each: specific gravity 1.003 to 1.030, pH about 4.5 to 8.0, and negative for glucose, protein, ketones, and blood. Beside every pad draw the nephron step that keeps that pad negative in a healthy person. One line: name the pad that turns positive first in uncontrolled diabetes, and the second one that follows it.

☐ **B** Draw three urine samples with their dipstick results filled in: one from an untreated type 1 diabetic, one from someone with glomerular damage, and one from someone who ran a hot marathon without drinking. Mark abnormal pads and specific gravity on each, and beside each draw the nephron arrow that failed or was pushed. One line: one of these three kidneys is entirely healthy, so name it and say what its dipstick is really reporting.

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COMP

Renal function calculation

You should be able to: Calculate glomerular filtration rate, clearance, and filtered load from a data set and interpret the values in a clinical case.

PICK ONE AND DO THAT ONE

☐ **A** Work a case on the page. Choose plausible values for plasma creatinine, urine creatinine, and urine flow rate, then draw the clearance calculation as a labeled flow of numbers with units carried through every step. Use your answer to calculate the filtered load of sodium at a plasma sodium of 140 mEq/L. One line: compare that filtered load against the sodium actually excreted in a day, and say what the gap tells you about the tubule.

☐ **B** Draw a two column comparison of two patients with the same urine output, one with plasma creatinine of 1.0 mg/dL and one at 4.0 mg/dL. Calculate each one's clearance and draw it as a bar against a marked normal near 125 mL/min. Then draw what the nephrons of the second patient are doing to produce that number. One line: say whether a fourfold rise in plasma creatinine means a fourfold fall in GFR, and why that relationship is not a straight line.

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21

Volume and osmolarity disturbances

COMP

You should be able to: Classify a disturbance as volume depletion or overload and as hypoosmotic, isosmotic, or hyperosmotic and predict the resulting fluid shift between compartments.

PICK ONE AND DO THAT ONE

☐ **A** Draw the fluid compartments as two boxes sharing a wall, intracellular at about two thirds of body water and extracellular at about one third, with height for volume and width for osmolarity starting near 290 mOsm/L. Redraw the pair for three cases: hemorrhage, heavy sweating without drinking, and swallowing several liters of plain water fast. Show which wall moved and which way water shifted. One line: name the case where cells swell, and the tissue where that is dangerous.

☐ **B** Draw a two by three grid of compartment pictures, volume depletion and volume overload down the side, hypoosmotic, isosmotic, and hyperosmotic across the top. Fill every cell with the two boxes drawn to the right shape, and write a real cause under each. One line: in exactly two cells the extracellular and intracellular volumes move in opposite directions, so name those two and say what drives water the way it goes.

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